# UNIT I

1. A discrete random variable X has the following probability distribution

| X | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| P(X=x) | a | 3a | 5a | 7a | 9a | 11a | 13a | 15a | 17a |

a) Find the value of a.

b) Find $P(X<3)$, $P(0<X<3)$, $P(X\geq 3)$

c) Find the distribution function of X.

d) $P(X>2/X<6)$

2. A random variable X has the following probability function:

| x: | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| P(x): | 0 | a | 2a | 2a | 3a | $a^2$ | $2a^2$ | $7a^2+a$ |

i) Find a

ii) Evaluate $P(X<6)$, $P(X\geq 6)$

iii) Evaluate $P(X<6/X\geq 4)$

iv) If $P(X\leq \lambda) > \frac{1}{2}$, find the minimum value of $\lambda$

3. A continuous random variable X has a pdf $f(x) = kx^2e^{-x}$, $x\geq 0$. Find k, $r^{th}$ moment about the origin and hence find the mean and variance

4. If the density function of a continuous random variable X is given f(x):

$$\begin{cases} ax & ; 0\leq x\leq 1 \\ a & ; 1\leq x\leq 2 \\ 3a-ax & ; 2\leq x\leq 3 \\ 0 & ; \text{otherwise} \end{cases}$$

i) Find the value of a.

ii) CDF of X.

5. Find the ~~Mean~~ MGF of binomial distribution and hence find the mean and variance.

6. State and prove Memoryless property of geometric distribution.

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Unit - II

1. The JPMF of $(X,Y)$ is given by $P(x,y) = k(2x+3y)$; $x = 0,1,2$; $y = 1,2,3$. Find all the marginal and conditional probability distributions. Also find the probability distribution of $(X+Y)$ and $P[X+Y \geq 3]$.

2. The Joint distribution of $X$ and $Y$ is given by $f(x,y) = \frac{x+y}{21}$, $x = 1,2,3$, $y = 1,2$. Find the marginal distribution. Also find $E[XY]$

3. The Joint pdf of $(X,Y)$ is given by $f(x,y) = 4xye^{-(x^2+y^2)}$, $x,y \geq 0$. Verify whether $X$ and $Y$ are independent or not

4. Two random variables $X$ and $Y$ have the joint density

$$f(x,y) = \begin{cases} 2-x-y, & 0<x<1,\ 0<y<1 \\ 0, & \text{otherwise} \end{cases}$$

Show that

$$\text{Cov}(X,Y) = -\frac{1}{144}$$

If $X$ and $Y$ are independent random variables with p.d.f $e^{-x}$, $x \geq 0$; $e^{-y}$, $y \geq 0$ respectively. Find the density function of $U = \frac{X}{X+Y}$ and $V = X+Y$. Are $U$ and $V$ independent?

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# UNIT - III

1. Show that the random variable process $X(t) = A\cos(\omega t + \theta)$ is a wide sense stationary if A and $\omega$ are constant and $\theta$ is uniformly distributed random variable in $(0, 2\pi)$

2. The transition probability matrix of the Markov chain $\{X_n\}$, $n = 1, 2, 3, \ldots$ having states 1, 2 and 3 is $P = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$ and the initial distribution is $P^{(0)} = (0.7, 0.2, 0.1)$. Find (i) $P(X_2 = 3)$ and (ii) $P(X_3 = 2, X_2 = 3, X_1 = 3, X_0 = 2)$.

3. Consider two random processes $X(t) = 3\cos(\omega t + \theta)$ and $Y(t) = 2\cos(\omega t + \theta - \pi/2)$, where $\theta$ is a random variable uniformly distributed in $(0, 2\pi)$. Prove that $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)\, R_{YY}(0)}$.

4. If the random process $X(t)$ takes the value $-1$ with probability $\frac{1}{3}$ and takes the value 1 with probability $\frac{2}{3}$. Find whether $X(t)$ is a stationary process or not.

5. The process $\{X(t)\}$ whose probability distribution under certain conditions is given by

$$P\{X(t) = n\} = \begin{cases} \dfrac{(at)^{n-1}}{(1+at)^{n+1}} & ; \ n = 1, 2, \ldots \\ at/1+at & ; \ n = 0. \end{cases}$$

Show that it is not stationary (or evolutionary).

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# UNIT IV

1. Two horses A and B were tested according to the time (in seconds) to run a particular race with the following results:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Horse A | 28 | 30 | 32 | 33 | 33 | 29 | 34 |
| Horse B | 29 | 30 | 30 | 24 | 27 | 29 | |

Test whether the horse A running faster than B at 5% level.

2. A random sample of 10 boys had the following IQ is 70, 120, 110, 101, 88, 83, 95, 97, 107, 100. Do these data support the assumption of a population mean IQ of 100? Find a reasonable range in which most of the mean IQ values of samples of 10 boys lie.

3. A sample of heights of 6400 Englishmen has a mean of 170 cm with a standard deviation of 6.4 cm while a sample of heights of 1600 Australians has standard deviation 6.3 cm with mean 172 cm. Do the data indicate that Americans are, on the average taller than the Englishmen at 1% level of significance?

4. A group of 10 rats fed on diet A and another group of 8 rats fed on diet B, recorded the following increase in weight.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Diet A | 5 | 6 | 8 | 1 | 12 | 4 | 3 | 9 | 6 | 10 |
| Diet B | 2 | 3 | 6 | 8 | 10 | 1 | 2 | 8 | | |

Find if the variances are significantly different.

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5. The following data are collected on two characters

| | Smokers | Non-Smokers |
|---|---|---|
| Literates | 83 | 57 |
| Illiterates | 45 | 68 |

Based on this, can you say that there is no relationship between smoking and literacy?

6. 4 Coins were tossed 160 times and the following results were obtained

| No. of heads : | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Observed frequencies : | 17 | 52 | 54 | 31 | 6 |

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# Unit - V

1. Analyze the variance in the following Latin - Square of yields of paddy where A, B, C, D denote the different methods of cultivation.

| | | | |
|---|---|---|---|
| D 122 | A 121 | C 123 | B 122 |
| B 124 | C 123 | A 122 | D 125 |
| A 120 | B 119 | D 120 | C 121 |
| C 122 | D 123 | B 121 | A 122 |

Examine whether the different methods of cultivation have given significantly different yields.

2. Analyse the variance in the following Latin square yields (in kgs) of paddy where A, B, C, D denote the different cultivation methods.

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| | | | |
|---|---|---|---|
| B 20 | C 17 | D 25 | A 34 |
| A 23 | D 21 | C 15 | B 24 |
| D 24 | A 26 | B 21 | C 19 |
| C 26 | B 23 | A 27 | D 22 |

Examine whether the different methods of cultivation have given significantly different in yields.

3. In order to compare three burners A, B & C an observation is made on each burner on three successive days. The data tabulated below.

| | A | B | C |
|---|---|---|---|
| Day-1 | 6 | 28 | 14 |
| Day-2 | 28 | 36 | 38 |
| Day-3 | 8 | 32 | 20 |

Perform an analysis of variance on these data and find whether the difference between (i) the days and (ii) the burner's significant at 5% level of significance.

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4. Five doctors each test five treatments for a certain disease and observe the number of days each patient takes to recover. The recovery times in days are as follows.

| Doctors | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| A | 10 | 14 | 23 | 19 | 20 |
| B | 11 | 15 | 24 | 17 | 21 |
| C | 9 | 11 | 20 | 16 | 19 |
| D | 8 | 13 | 17 | 17 | 20 |
| E | 12 | 15 | 19 | 15 | 22 |

Is there any significant difference between the doctors and their treatments?

5. There are three main brands of a certain powder. A set of 120 sample value is examined and found to be allocated among four groups (A, B, C & D) are three brands (I, II & III) as shown here under

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| | | Groups | | | |
|---|---|---|---|---|---|
| | | A | B | C | D |
| Brands | 1 | 0 | 4 | 8 | 15 |
| | 2 | 5 | 8 | 13 | 6 |
| | 3 | 18 | 19 | 11 | 13 |

Is there any significant difference in brand preference? Answer at 5% level of significance using one-way ANOVA?

b. The following table shows the lives in hours of 4 brands of electric lamps.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| A: | 1610 | 1610 | 1650 | 1680 | 1700 | 1720 | 1800 | - |
| B: | 1580 | 1640 | 1640 | 1700 | 1750 | - | - | - |
| C: | 1460 | 1550 | 1600 | 1620 | 1640 | 1660 | 1740 | 1820 |
| D: | 1510 | 1520 | 1530 | 1570 | 1600 | 1680 | - | - |

Perform an analysis of variance test the homogenity of the mean lives of the 4 brands of lamps.

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